

SECTION B

6 (short)

- (a) Prove that for any integer $n > 0$ the Boolean expression below is true.

$$A_1 + A_2 + \dots + A_n = \overline{\overline{A_1} \cdot \overline{A_2} \cdot \dots \cdot \overline{A_n}}$$

[5]

- (b) Draw a circuit diagram that implements the Boolean expression

$$A.B + C$$

by means of NOR gates only, using the least number of gates possible.

[5]